

Situational Anomaly Detection in Multimedia Data under Concept Drift

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ABSTRACT

Anomaly detection has been a very challenging and active area of research for decades, particularly for video surveillance. However, most of the works detect predefined anomaly classes using static models. These frameworks have limited applicability for real-life surveillance where the data have concept drift. Under concept drift, the distribution of both normal and anomaly classes changes over time. An event may change its class from anomaly to normal or vice-versa. The non-adaptive frameworks do not handle this drift. Additionally, the focus has been on detecting local anomalies, such as a region of an image. In contrast, in CCTV-based monitoring, flagging unseen anomalous situations can be of greater interest. Utilizing multiple sensory information for anomaly detection has also received less attention. This extended abstract discusses these gaps and possible solutions.

CCS CONCEPTS

• **Computing methodologies** → *Modeling methodologies*.

KEYWORDS

multimodal anomaly detection; long term surveillance; unsupervised modeling; concept drift

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1 MOTIVATION

Anomaly detection refers to the task of finding novel observations or patterns in data [4]. The detection of abnormal observations is of great interest in a variety of fields, including economics, medicine, surveillance, data mining, and robotics. There are multiple challenges involved in anomaly detection for audio-visual surveillance, such as scarcity of natural anomaly samples, high intra-class variance, complex Spatio-temporal context associated with events, and the presence of concept drift. In order to deal with the lack of natural anomaly samples and the high intra-class variance, the research

has been shifted from supervised machine learning frameworks to unsupervised deep learning frameworks.

The challenges that I am targeting are discussed below:

Temporal width of events: To include local Spatio-temporal context, state-of-the-art works consider 3D patches extracted from video chunks of 5-10 frames. The patches are also regarded as events in these works. The attention on variable event size in spatial dimension is taken care of by sampling patches on different frame resolutions. However, in real life, the event may be of variable temporal context too, i.e., the size of an event may vary on the temporal dimension. Under-sampling may result in no detection of events, whereas oversampling may cause the inclusion of multiple events. It becomes even more challenging where there are multiple events occurring concurrently in the scene.

Adaptive learning: The state-of-the-art works claim to include global Spatio-temporal context by using the whole training data at once for normality estimation. However, the training data spans just a few minutes of video data. A model trained on few minutes of data may not perform well for long-term surveillance. Let's consider an example of a running person in the UCSD Peds dataset [25]; the running person in a pedestrian area is rare and hence regarded as an anomaly there; what if that situation is followed by a marathon in which athletes are passing from the pedestrian area. Initially, this situation is anomalous, whereas after watching enough samples of athletes running, the situation becomes new normal. However, these models will always flag the running event as abnormal because it was regarded as an anomaly during model training. The class of an event changes due to drift in underlying data distribution over time in dynamically changing environments [39], also regarded as concept drift. Consequently, the model should continuously adapt its notion of anomaly. It should remember the event for a longer period while gradually forgetting prior occurrences in order to place a greater emphasis on recent trends in event statistics. This concept of memorizing is analogous to how the human brain operates; we tend to analyze an object/event based on current observations and gradually forget the extremely distant ones.

Multimodal information-based anomaly modeling: Another limitation of the state-of-the-art works is that only a single source of information, either video or audio, is considered for anomaly modeling. Relying solely on visual data implies a number of assumptions about the CCTV cameras' uninterrupted and robust performance. Besides this, some anomalous situations may go undetected if we consider only audio or video from the target scene. We are more accurate in situation analysis when additional sensory data such as audio, video, pressure, and temperature are provided [4, 7]. In the monitoring task, the audio and video together provide complementary information for comprehending the scene.

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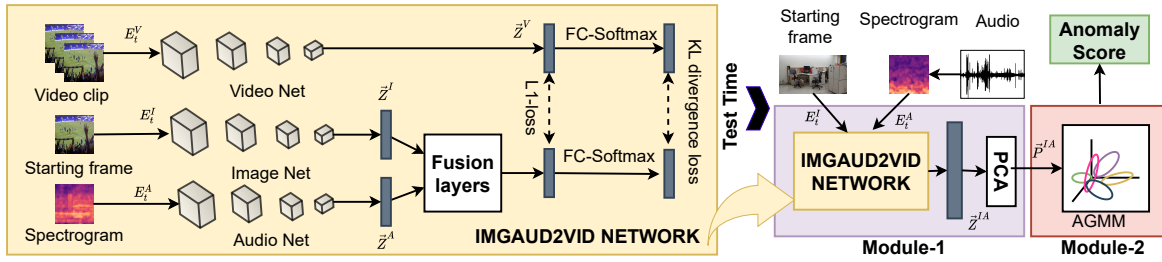


Figure 1: Overview of the proposed framework in Work-2. Module-1 generates deep multimodal scene representation, and Module-2 is used for adaptive scene learning.

Also, I found that most of the existing public audio/video datasets have a short duration of clips (UCSD Peds [25], AVENUE [23]) or contains samples from stationary classes (DCASE [26], Subway [1], UCF-Crime [33]). The dataset to be useful for evaluating adaptive anomaly detection framework should be sufficiently long to have concept drift. Also, the footage should be recorded in one place and untrimmed. Seeing the lack of such datasets, I collected multiple long-duration (ranging from 1.5 to 10 hours) datasets.

1.1 Novelty

My ongoing research work on anomaly modeling has the following novelty and significance relative to the state-of-the-art works.

- The study exploits unsupervised adaptive anomaly modeling that works better than non-adaptive approaches for detecting anomalies in the presence of concept drift.
- A dynamic Huffman coding-based anomaly detection framework is developed that follows a learning and remembering strategy to handle long-term temporal context and, hence, performs better than the Adaptive Gaussian Mixture Model.

2 BACKGROUND AND RELATED WORK

Unsupervised Anomaly Detection: Anomaly detection in the video has been extensively studied in past decades. Research has transitioned from initial local Spatio-temporal hand-crafted features-based supervised methods to mid-level descriptor-based unsupervised approaches [21, 37]. State-of-the-art unsupervised anomaly detection techniques are predominantly based on applying reconstruction-based [14, 37], prediction-based [38, 40], or hybrid [21, 41] deep generative frameworks. However, the data used to train these models is too short and discontinuous to handle concept drift events. These models will always classify an event as abnormal during the test phase if it was regarded as abnormal during the training phase, regardless of how frequently it occurs in the distant future. Abnormality detection in the audio modality has primarily focused on detecting fixed kinds of anomaly [3, 5, 8, 10, 11, 30] (e.g., glass break, gunshot, vehicle crashes, shouting, etc.). Detecting anomalies in data with concept drift is still a challenging task.

Multimodal Anomaly Detection: Few works have utilized time-series multimodal information for anomaly detection. HMM and LSTM based static classifiers were used to model the normal behavior of a PR2 robot using haptic, visual, auditory, and kinetics information [28, 29]. The deviation from normal behavior was flagged by

detecting a reduction in log-likelihood or an increase in reconstruction loss. Kitller et al. [17] detected potential anomalies on the basis of incongruence between HMM-based separate audio and video models. Kuklyte et al. [18, 19] used a q-mode ($q=100$) clustering technique to build a normality model on low-level features from audio and visual modalities. The aforementioned works employ static learning and evaluate their approaches using datasets having few simple indoor activities, making them unsuitable for detecting anomalies under concept drift. Moreover, the early fusion or late fusion strategies have been considered in such works, whereas an alternative approach, intermediate or hybrid fusion, has been preferred by the researchers lately.

3 APPROACH

The purpose of this section is to discuss the work that has been done so far toward the goal of developing a multimodal adaptive framework for scene-level anomaly detection. They are discussed below.

Work-1: Unsupervised adaptive anomaly modeling: Here, the focus was on developing an unsupervised adaptive anomaly detection framework for the video stream. Inspired from the successful deployment of univariate Adaptive Gaussian Mixture Model (AGMM) for various tasks such as foreground-background separation [9, 32], event detection [6, 7, 27], outlier detection in non-multimedia data [15, 36], I explored multivariate AGMM for scene level anomaly modeling. It facilitates an unsupervised adaptive learning. There can be a maximum of K modes (Gaussian) in the mixture. Each mode represents a unique event. For each mode, mean and variance are maintained. Additionally, weight is stored to account for the estimated frequency of that particular event. These values are dynamically updated upon the arrival of each new sample using an online K-means approximation strategy. If a new sample matches any existing mode, then the mean and variance of that mode are updated; otherwise, a new mode is created. Further, if the number of modes exceeds K , the weakest mode is replaced with a new one; otherwise, a new mode is simply added to the mixture. A mode with a higher weight indicates that the associated event occurred more frequently, and thus events matching this mode receive a low anomaly score and vice versa. I experimented on 10 hours of untrimmed video dataset. A combination of a semantic and motion-based low-level feature was used. The detailed approach and results are reported in the following publications: *Kumari P, Saini M. Multivariate Adaptive Gaussian Mixture for Scene Level Anomaly Modeling. In 2020 IEEE Sixth International Conference on*

Multimedia Big Data (BigMM) 2020 Sep 24 (pp. 54-62). IEEE.

Work-2: Unsupervised adaptive anomaly modeling with multimodal data: Here, I explored multimodal information-based anomaly modeling. Joint audio-visual frameworks have proven to be more effective than video-only frameworks in a variety of applications such as video categorization [16, 34], identity verification [22, 31], activity classification [2, 12], etc. Inspired by the efficacy of multimodal approaches in a variety of applications; I propose a joint audio-visual anomaly detection framework (shown in Figure 1). The framework constitutes two modules, namely Module-1 and Module-2. In Module-1, a teacher-student network is used that hallucinates a video motion descriptor from a video frame and the accompanying audio. Specifically, the IMGAUD2VID network [12] with some customization in the architecture is adopted. The network is aimed at establishing a binding of audio and visual modalities so that correlated audio and video cues can be aggregated. In the architecture, ‘Video Net’ represents the teacher module; ‘ImageNet’ and ‘Audio-Net’ constitute the student network. The teacher network is a pre-trained model on the video modality, which distills its information on the student network. Since just a single image and the accompanying audio is used for feature extraction at the test time, the joint descriptor is low-cost too. Further, to reduce the dimensionality of extracted deep features, a Principal Component Analysis (PCA) [35] is applied. Module-2 learns the distribution of conceptually unique events evolving in the target scene. We utilize our AGMM based framework discussed in Work-1 as a baseline for learning the scene dynamics. To the best of our knowledge, there are no publicly available datasets for the evaluation of multimodal adaptive anomaly detection frameworks. Therefore, two datasets are collected, and the proposed framework is compared against uni-modal frameworks and a non-adaptive framework on these datasets. The associated paper is under review in ICVGIP 2021.

Work-3: Unsupervised adaptive anomaly modeling to handle longer temporal context: While AGMM performs well under concept drift, particularly when adapting to the trend in the data, it may miss transitions from abnormal to normal for some events that re-appears after longer time spans. This is especially true when there are too many different types of events, and the scene dynamics changes rapidly. The static model memory of K (3-10) Gaussians will not be sufficient to handle it. Towards this, I proposed a dynamic Huffman coding-based anomaly modeling framework for streaming audio data (Figure 2). Dynamic Huffman coding allows encoding without knowledge of the source distribution. However number of unique events (K) is still required. I proposed an extension of dynamic Huffman coding in which this K does not need to be specified in advance, and new nodes can be added freely. I experimentally found that the number of nodes does not explode no matter how random data is taken. Also, a modified tree reorganization that allows a variable amount of weight update is devised (in the case of standard dynamic Huffman coding, a constant weight update is followed). Here, unique events are stored in tree nodes with mean vector and weight information. Nodes having higher weight are stored relatively upper part of the tree, and hence the coding cost will be low for them. Hence relative coding cost measure is taken as an anomaly score. Also, to minimize falsely initialized nodes (often the case in a fast-changing environment [27]), a node merge scheme is devised. Nodes that are too close are iteratively

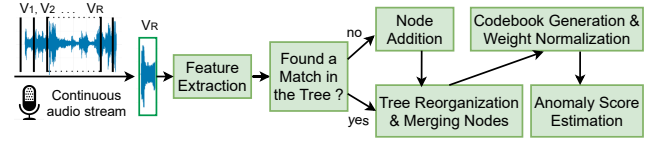


Figure 2: Coarse level architecture of the proposed framework in Work-3.

Table 1: Datasets: ‘TV Series’ and ‘12 Angry Men’ datasets are taken from the web, and the rest others are self-collected.

Dataset Name	Modality	Length	Purpose
Lab-1	Audio	10 hours	No. of Nodes
Reception Area	Audio	10 hours	No. of Nodes
Outdoor Canteen-1	Audio	10 hours	No. of Nodes
TV Series [13]	Audio	10 hours	No. of Nodes
Lab-2	Audio	1.33 hours	Adaptiveness
12 Angry Men [24]	Audio	1.5 hours	Performance
Outdoor Canteen-2	Audio	3 hours	Performance
An Outdoor Area	Audio	3 hours	Performance
Daily-Life	video	10 hours	Performance
Phd-Lab	audio-visual	82 minutes	Performance
Reception	audio-visual	189 minutes	Performance

Table 2: Work-2: Performance comparison of the proposed approach, it’s uni-modality variants, handcrafted video feature based AGMM [20], and a non-adaptive approach [37].

Method →	Proposed			[20]	[37]
Modality →	IA	I	A	V	IA
Datasets ↓					
PhD-Lab	73.11	71.67	53.71	63.92	58.67
Reception	79.22	64.38	53.76	67.35	74.26

merged. It also reduces the memory requirement. Here again, there is a lack of suitable audio datasets for long-term anomaly modeling. Therefore multiple indoor and outdoor datasets were collected for evaluation. The work is under submission with two major revisions in ACM Transaction on Multimedia. It is available at arXiv: <https://arxiv.org/abs/2102.10515v2>.

4 RESULTS

The list of collected datasets is given in Table 1. The top five datasets are not labeled (not required); the rest six datasets are labeled manually with the help of three security persons.

Work-1 is evaluated on the ‘Daily-Life’ video dataset; some results from my paper [20] are shown in Figure 3. Experiments with different event sizes show that the model attains the best performance with an event size of 2 seconds for this dataset (Figure 3a). Similar AUC (Area Under the ROC Curve) for each 1 hour of the data shows that the performance does not degrade over time (Figure 3b). In Figure 3c, a comparison with two non-adaptive approaches is

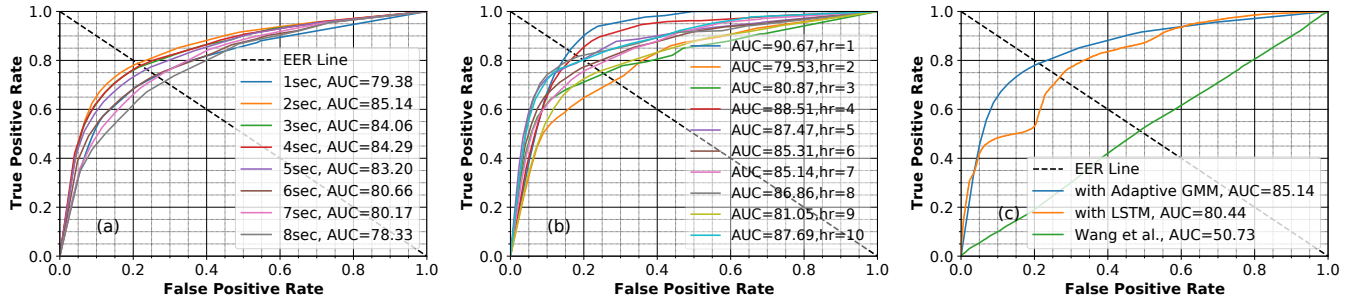


Figure 3: Work-1: Fig. (a) choosing best suitable event size, Fig. (b) showing performance for each hour of data, and Fig. (c) show performance comparison of our approach with non-adaptive approaches [37] and a baseline auto-encoder based LSTM

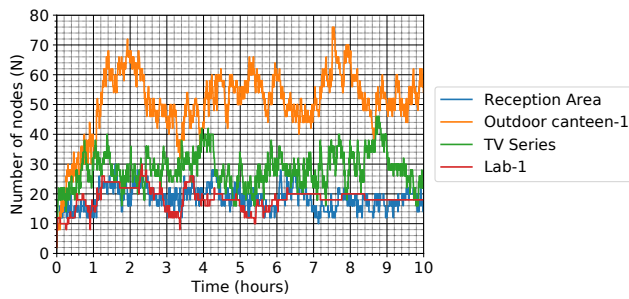


Figure 4: Work-3: Number of nodes in the tree over a duration of 10 hours in multiple challenging datasets.

plotted. The First 1 hour of data is used for training these non-adaptive approaches. Reported AUCs are for the whole 10 hours of data. We can see that for non-adaptive approaches, performance is very low compared to the proposed adaptive framework.

For evaluating joint feature-based adaptive anomaly detection framework, two datasets, namely ‘PhD-Lab’ (82 minutes) and ‘Reception’ (189 minutes), are collected (Work-2). AUC comparison of the proposed approach with its unimodal variants, handcrafted video feature-based AGMM [20], and a non-adaptive approach (with the proposed joint feature) [37] are shown in Table 2. We can clearly see that joint feature-based adaptive modeling outperforms unimodal feature-based anomaly modeling.

In Work-3, the number of nodes in the dynamic Huffman tree over time is shown in Figure 4 for four 10 hours of audio datasets (‘Lab-1’, ‘Reception Area’, ‘Outdoor Canteen-1’, and ‘TV-Series’). We can see that number of nodes does not explode with an unconstrained number of nodes strategy too. In Figure 5, a comparison of baseline AGMM and proposed dynamic Huffman coding-based anomaly modeling on three audio datasets (‘Outdoor Canteen-2’, ‘An Outdoor Area’, and ‘12 Angry Men’) is given. We can see that the proposed approach outperforms baseline AGMM with a gain of 2-8% AUC. The improvement is due to the fact that no replacement of weaker cluster in the proposed framework trigger very slow forgetting of the past compared with baseline AGMM.

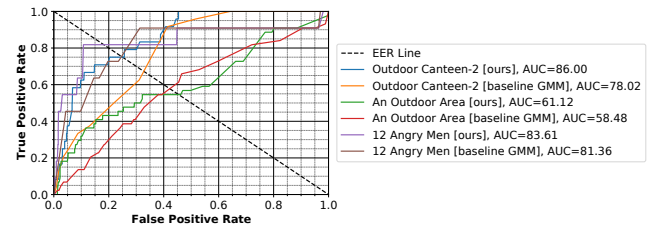


Figure 5: Work3: ROC plots for the proposed approach and AGMM on selected hyper-parameters.

5 WORK IN PROGRESS

In all the works, a static event duration is assumed. However, the assumption does not hold true for real-life scenarios. The proposed approaches will most likely perform better after incorporating adaptive event size. So, currently, I am focusing on variable event modeling for long-term anomaly modeling.

6 IMPLICATIONS AND REPRODUCIBILITY

There are no ethical implications of the work; passive video-based monitoring is already there. The proposed framework can bring huge assistance to security operators for the automated surveillance and hence will facilitate active surveillance. Scene-level adaptive anomaly modeling has many other potential usages. Apart from public safety, the model can be applied in various applications such as patient monitoring, elderly care, depression detection, disease outburst prediction, farm surveillance, etc. The model can be used for long-term monitoring without degradation in performance.

The proposed datasets will be available at the drive link¹. The code to reproduce the results will be shared at the GitHub account².

7 ACKNOWLEDGEMENTS

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¹Dataset Link

²GitHub account

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